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Combine intra- and inter-flow: A multimodal encrypted traffic classification model driven by diverse features

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ABSTRACT

The increasing prevalence of encryption for data security in network communication highlights the urgent need for effective encrypted traffic identification methods. Challenges such as payload concealment, diverse encryption protocols, and evasion tactics necessitate innovative approaches, including deep learning and multiple features integration. Multimodal learning, which utilizes features from multiple sources, proves more effective than single-modality methods, offering a more comprehensive analysis. However, current methods often overlook valuable inter-flow information, hindering optimal classification. Hence, we propose a novel multimodal encrypted traffic classification model driven by diverse features, named MeDF, which combines intra- and inter-flow features. MeDF leverages both intra-flow and inter-flow features: it constructs spectrograms of flows' raw bytes and extracts statistical features, combining the two as the intra-flow features. MeDF also builds flow relation graph to extracts inter-flow features, helping the model learn the complex relationships between flows. The intra- and inter-flow features complement each other and extract the maximum amount of valid information from encrypted traffic. Thus, MeDF overcomes the performance limitations of existing multimodal models for encrypted traffic classification. MeDF is validated on 2 real-world traffic datasets with accuracy of 98.57% and 94.73%. The outperformance can be attributed to the complementary information of the employed features and the modeling of complex relationships, when compared to both classical single-modality classification methods and state-of-the-art multimodal methods.

1. Introduction

Network traffic classification is a crucial tool for network supervision [1,2]. In recent years, machine learning, particularly deep learning methods capable of handling intricate data relationships and large-scale data processing, has gained more preference over traditional methods [3,4].

In order to fully unleash the potential of the model, it is typically necessary to choose appropriate features from encrypted traffic data for learning. We have summarized the types of features used in recent network traffic classification works in Table 1. These features include statistical feature, Temporal feature, original traffic and graph feature.

Statistical features of network flows, such as packet number, average packet size, and average arrival time, prompt the use of learning-based algorithms to model feature distributions for network flow classification [5–7]. Temporal features include the packet length sequence and packet arrival interval sequence, treating flows as time series [8]. Some models use raw flow data, like L4 payload, as input, integrating feature/representation learning and classification into a unified pipeline [6,7,9]. Handling these features typically involves the use of CNN, MLP, ensemble learning, and Transformer models.

Additionally, when different network flows share common attributes like IP, port, and protocol, these correlation features contribute to effective traffic classification. Often represented through graph structures, these are referred to as graph features. It is crucial to note that certain studies, despite using graph structures, represent an individual flow with a single graph, deviating from the conventional understanding of graph features in this context [10–15]. GNN is commonly used to learn graph feature.

It can be observed that the majority of methods for encrypted traffic classification typically focus solely on one aspect of encrypted traffic features [16,17]. This limitation hinders the full utilization of information from other aspects during classification, thereby weakening overall

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Table 1
Summary of the features used in recent network traffic classification work and our MeDF.

Model \ Feature	Intra-flow feature			Inter-flow feature
	Statistical feature	Temporal feature	Original traffic	Graph feature
1D-CNN [9]	✗	✗	✓	✗
XGBoost [5]	✓	✗	✗	✗
ACID [8]	✗	✓	✗	✗
ProGraph [13]	✗	✗	✗	✓
WTAGraph [15]	✗	✗	✗	✓
AppNet [6]	✓	✗	✓	✗
MIMETIC [7]	✓	✗	✓	✗
MeDF (Ours)	✓	✓	✓	✓

effectiveness. To address this, multimodal learning methods have been introduced to the task of encrypted traffic classification. [6,7]

Multimodal learning, a technique integrating information from two or more modalities, facilitates effective classification and prediction [18]. In contrast to general encrypted traffic classification methods, approaches based on multimodal learning can simultaneously leverage features from different perspectives of traffic, thereby enhancing classification effectiveness. These approaches treat features as different modalities and learn them using an appropriate model to combine multiple sources of information for classifying encrypted traffic.

Network traffic inherently consists of multimodal data with diverse features. The features fall into two main categories: intra-flow features and inter-flow features. Intra-flow features consist of statistical characteristics of the traffic, raw bytes, etc. These features can be directly quantified and represented in Euclidean space, constituting Euclidean space features. They help capture the fundamental patterns and behaviors of the traffic, providing an intuitive feature representation for classification tasks. Inter-flow features can be extracted from the structural or sequential characteristics of encrypted traffic. For instance, relationships between flows in the traffic can be captured through graph representation learning methods, which are more naturally represented in non-Euclidean spaces such as graph space, constituting non-Euclidean space features. The diverse features enable multimodal learning to be effectively applied to encrypted traffic classification tasks.

Currently, many research studies focus on either intra-flow or inter-flow features exclusively. Our focus, however, lies on methods employing multimodal learning. Theoretically, multimodal learning necessitates complementary and heterogeneous characteristics among different modalities. Utilizing features meeting these criteria as modalities enhances the information exchange between them, thereby improving the effectiveness of multimodal learning. Despite the utilization of multimodal learning in existing research, feature selection remains limited to a single type of intra-flow feature. This constraint may hinder the model's ability to fully capture features from diverse perspectives of network flows, potentially impacting classification performance.

Combining intra-flow and inter-flow features as Euclidean space and non Euclidean space features respectively has certain advantages:

1. Complementarity: Euclidean space features are typically suitable for capturing linear relationships and regular geometric structures, while non Euclidean space features excel at handling complex topological structures and nonlinear relationships. This complementarity can enable the model to comprehensively understand and characterize encrypted traffic, and also meet the requirements of multimodal learning.
2. Enhanced representation capability: By combining two spatial features, the model can more flexibly represent different types of data and relationships. This enhanced representation ability helps to improve the generalization ability and accuracy of the model.
3. Improving the generalization ability of the model: Non Euclidean features can help the model capture deeper, non explicit data structures and relationships, while Euclidean features can provide more intuitive and conventional data representations. This diversified feature representation method helps to improve the model's adaptability and generalization ability to new situations, making the model more robust when encountering new data.

In this study, we introduce MeDF, a multimodal encrypted traffic classification model, driven by a diverse array of features. Our approach emphasizes the simultaneous utilization of both intra-flow and inter-flow features. Specifically, we employ statistical features and time–frequency domain features as intra-flow features for network flows, while graph features serve as inter-flow features. Given that traffic data inherently encompasses temporal features in both the time and frequency domains, we treat these features as intra-flow components within the Euclidean space. To comprehensively characterize features in both domains, we leverage spectrograms, allowing us to address the potential oversight of using only one aspect of the features and mitigating the lack of information on the other.

Recognizing the necessity of non-Euclidean features, we introduce the concept of a relation graph of flows to represent inter-flow features. This supplementary representation in non-Euclidean space enhances the model's capacity to capture diverse aspects of network traffic relationships. Our classification process entails the use of three distinct deep learning models, each specialized in learning one of the three feature types. Subsequently, these models seamlessly integrate to achieve a comprehensive classification outcome. This holistic approach, combining Euclidean and non-Euclidean spaces, ensures our model effectively addresses the intricate challenges associated with encrypted traffic classification.

To the best of our knowledge, this is the first encrypted traffic classification model that uses both of intra-flow and inter-flow features. The main contributions of this study are as follows.

1. We construct a multimodal classification model MeDF. It can make full use of the comprehensive information extracted from inter-flow and intra-flow features, meeting the needs of multimodal learning for feature complementarity and heterogeneity.
2. We use the flow relation graph to characterize the inter-flow features while spectrogram and statistical feature to characterize the intra-flow features. These complementary features on both aspects can provide more comprehensive information about encrypted traffic to the model. Therefore, MeDF can achieve better classification result.
3. We validate the model on the real-world dataset and achieve good results, proving that classification using both intra-flow features and inter-flow features is highly effective.

The subsequent chapters of this paper are organized as follows: Section 2 focuses on the works related to network traffic classification and some basic knowledge required in this paper. Section 3 explains the model framework of MeDF. Section 4 verifies the validity of the model through experiments. Section 5 concludes the work of this paper.

2. Background and related work

Network traffic classification is one of the key research problems in the field of network security. This section focuses on a review of existing work, including the definition of network traffic classification (Section 2.1), feature-based classification methods (Section 2.2), and current works related to encrypted traffic classification that use multimodal learning (Section 2.3).

2.1. Definition of network traffic classification

Encrypted traffic classification refers to the identification of the category to which the traffic belongs without decrypting the traffic, such as the application generating the traffic or the purpose of the traffic. The encrypted traffic classification is typically done at the packet, flow, or session level. A flow is a set of packets represented by a five-tuple, including destination IP, source IP, destination port, source port, and protocol. A session refers to the bidirectional flow between two network nodes over a period of time.

Assuming we have a set of encrypted traffic data $\mathcal{D} = (\mathbf{x}_i, y_i), i = 1, \ldots, N$, where N is the number of samples, $\mathbf{x}_i$ represents the feature representation of the ith sample, and y_i is the corresponding class label.

In multimodal learning, $\mathbf{x}_i$ can be composed of data from multiple modalities, such as raw bytes, packet size distributions, etc. Therefore, $\mathbf{x}_i$ can be further represented as $\mathbf{x}_i = (\mathbf{x}_i^1, \mathbf{x}_i^2, \ldots, \mathbf{x}_i^M)$, where M is the number of modalities, and $\mathbf{x}_i^m$ represents the feature representation of the ith sample in the mth modality.

The structure of a multimodal learning model typically includes submodels that process features from each modality and a fusion network that combines these features for final classification. In the context of encrypted traffic classification, this can be represented as follows:

Modality feature extraction: For each modality m, there exists a function $f^m(\cdot)$ to extract relevant features, i.e., $\mathbf{z}_i^m = f^m(\mathbf{x}_i^m)$.

Modality fusion: After extracting the features$\mathbf{z}_i^1, \mathbf{z}_i^2, \ldots, \mathbf{z}_i^M$, these features are combined using a fusion function$f(\cdot)$, i.e., $\mathbf{z}_i = f(\mathbf{z}_i^1, \mathbf{z}_i^2, \ldots, \mathbf{z}_i^M)$.

Classification: Finally, a classification function $h(\cdot)$ is used to predict the class from the fused features, i.e., $\hat{y}_i = h(\mathbf{z}_i)$.

2.2. Feature-based methods

The primary concept of feature-based classification methods is to extract various types of features of network traffic and combine them with classification algorithms, such as machine learning or deep learning, to accomplish the classification of traffic. The existing feature-based work can be categorized into two categories: methods based on intra-flow features and methods based on inter-flow features.

2.2.1. Intra-flow-feature-based methods

The methods based on intra-flow features can be categorized into attribute-based features and sequence-based features. Among them, attribute-based features include statistical features such as maximum packet size, the average packet size of packets and string features extracted directly from the flow.

Machine learning or deep learning algorithms can use these features as input to classify encrypted traffic. Taylor et al. [19]. designed burst and flow statistical features and proposed AppScanner for fingerprint mobile applications. Chen et al. [20]. proposed Multi-Attribute Association Fingerprinting (MAAF) to classify mobile encrypted services. MAAF uses domain names, certificates, and data length to predict the application to which the encrypted flow belongs.

The sequence-based features of encrypted traffic include packet length and message state sequences. Liu et al. [21]. used first-order Markov chains to model packet length sequences and constructed a multi-attribute Markov probabilistic fingerprint model MaMPF. Liu et al. [22]. also designed a neural network model called Flowing Sequence Network (FS-Net), which consists of an encoder, decoder, and classifier. The encoder converts the input length sequence into encoded features, while the decoder attempts to recover the sequence by reconstructing the layers; the classifier uses the encoded features to identify the application class of the flow.

In our model, We use the spectrograms of flows as the representation of intra-flow features. Spectrograms can provide both time domain and frequency domain attributes of flows, avoiding the lack of information due to the use of only a single attribute. In addition, we also utilize statistical features as supplementary information to ensure that the model captures sufficient information from the traffic.

2.2.2. Inter-flow-feature-based methods

Inter-flow features mainly refer to the relationship features in network flows. Different network flows may have the same source/destination IP, protocols, and representative work to construct flow relation graph, and train GNN to complete the classification of encrypted traffic.

Yang et al. [15] propose WTAGRAPH, a network tracking and ad detection framework based on Graph Neural Networks (GNN). We first construct an Attribute Homogeneous Multigraph (AHMG) representing HTTP network traffic, and formulate network tracking and ad detection as tasks of edge representation learning and classification based on GNN in AHMG. Li et al. [13] proposed ProGraph,a graph-propagation-based method. It builds a correlation graph with session clusters aggregated from different networks, upon which effective graph propagation can be implemented iteratively to predict labels for test nodes. Cai et al. [23] proposed the MEMG method, which combines Markov chains and graph neural networks to represent the encrypted traffic of mobile applications as a first-order Markov chain graph representation, transforming the encrypted traffic classification problem into a graph classification problem.

Essentially, intra-flow features are features in Euclidean space, which are the various properties of the encrypted flows themselves. Meanwhile, inter-flow features are features on non-Euclidean space, which can reflect the relationship between multiple network flows. Our model combines these two types of features to classify encrypted traffic. It can preserve the complete information of encrypted traffic and optimize the classification effect.

2.3. Multimodal-learning-based methods

Multimodal learning refers to the process of integrating information from several modalities, specifically, combining information from multiple perspectives of an object, to perform classification. When performing classification, a single modality typically cannot contain all the valid information needed to produce accurate prediction results.

In contrast to from the general feature-based encrypted traffic classification methods, the approach of multimodal learning in encrypted traffic classification aims to synthesize multiple features of traffic by leveraging the concept of multimodal learning. The approach treats different features of traffic as different modalities, and learn them using models that are compatible with the modalities to unify multifaceted information to complete the classification of encrypted traffic. Aceto et al. proposed MIMETIC [7], a multimodal deep learning framework that uses the first 576 bytes of payload and four protocol features as input sources. The payload part is processed using a one-dimensional CNN and the protocol feature part is processed using a lightweight RNN structure GRU. Most previous approaches work by obtaining the payload portion of the first packet or by piecing together a mixed payload by extracting specific bytes from multiple packets, which lacks feature information of the entire network flow.

Wang et al. proposed a multimodal encrypted traffic classification framework, called AppNet. The framework uses a 1-D CNN to extract features from the first 1014 bytes and uses LSTM to learn the temporal relationship of the packet length sequence, and finally connects the features learned from both perspectives for classification [9]. Lin et al. proposed PEAN [24], a model that takes raw bytes and length sequences as input. It utilizes Transformer and bidirectional LSTM as sub-models and employs self-attention mechanism to learn deep relationships between network packets.

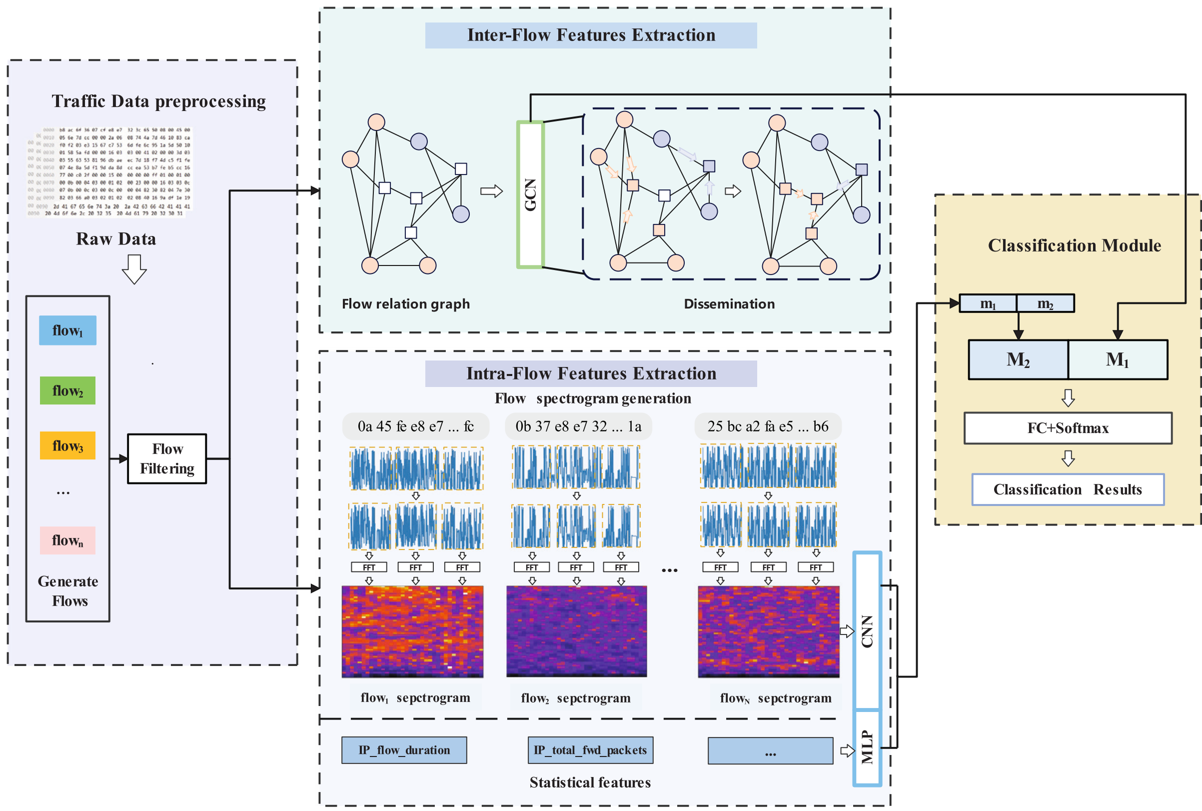


Fig. 1. The framework of MeDF.

Numerous current encrypted traffic classification methods using the multimodal learning use a combination of multiple intra-flow features. Additionally there is a lack of research on using both intra-flow features and inter-flow features for encrypted traffic classification. Therefore, in this study, we consider using the time–frequency domain features of flows as intra-flow features and the flow relation graph as inter-flow features of encrypted traffic.

3. Description of MeDF

This section introduces MeDF, the proposed encrypted traffic classification model. This model mainly consists of four parts: Traffic data preprocessing, Inter-flow features extraction, Intra-flow features extraction and classification module, as shown in Fig. 1. It aims to achieve the integration of intra-flow and inter-flow features to meet the requirements of multimodal learning for modality complementarity and heterogeneity. To achieve this, we design two modules for extracting preprocessed traffic features: one for intra-flow features and the other for inter-flow features.

Specifically, we construct a flow relationship graph to reflect the connections between different flows and utilize graph convolutional networks to extract inter-flow features. For intra-flow feature extraction, it involves two aspects: flow spectrogram and flow statistical features. Since traffic can be considered a type of time series data, we aim to use spectrograms to simultaneously capture its features in both the time and frequency domains. Additionally, combining statistical features, known for their universality and effectiveness, forms the intra-flow features. Finally, we input both inter-flow and intra-flow features into the classification module to complete the final classification. We describe each module in detail as following.

3.1. Traffic data preprocessing

The pcap data is divided into flows based on five tuples {source IP, source port, destination IP address, destination port, protocol}. It is also necessary to filter out the following three types of traffic.

Algorithm 1: Framework of MeDF

Input: Several pcap files containing m flows in total
Output: Classification results

Initialize M as an empty set;
Let $i = m$;
Intra-Flow features extraction:
while $i > 0$ **do**
  Extract the raw byte sequence $\mathbf{RBS_i}$ of $\mathbf{flow_i}$;
  Generate the spectrogram $\mathbf{spec_i}$ of $\mathbf{RBS_i}$;
  Extract the statistical features vector $\mathbf{sfv_i}$ of $\mathbf{flow_i}$;
  Input $\mathbf{spec_i}$ and $\mathbf{sfv_i}$ into CNN and MLP respectively, and obtain sub-modalities m_{1_i} and m_{2_i};
  Let $M_{1_i} = f(m_{1_i}, m_{2_i})$,where f is the modality fusion function;
  $i = i - 1$;
Inter-Flow features extraction:
Build the relation graph G of all the m flows;
Input G into GCN;
while $i > 0$ **do**
  Obtain the representation vector of the node i as the modality M_{2_i};
  $i = i - 1$;
Multimodal classification:
while $i > 0$ **do**
  Let $M_i = f(M_{1_i}, M_{2_i})$;
  Save M_i to M;
  $i = i - 1$;
Input M into the classification model to obtain the *result*
return *result*

1. **TCP Handshake Failed flows**. The data packets of a failed handshake do not carry with any valid application information and does not provide valid information for the actual service traffic identification. The TCP handshake failure packet does not carry any valid application information and does not provide valid information for actual service traffic identification. The filtering criteria for this type of session is that the transport layer protocol is TCP and the entire session handshake information is incomplete or there is no service load.
2. **DNS domain name query flows**. The IP address and port of the DNS server are generally different from the IP address and destination port of the specific business traffic. Therefore, even if a traffic service requires the use of a DNS service, its DNS query traffic and the business-specific traffic will be divided into different sessions when session segmentation is performed. Separate DNS traffic sessions are less helpful for traffic type identification. The traffic filtering criteria for this type of session is that the entire session data packet is a DNS protocol packet.
3. **LLMNR protocol flows**. Application Scenarios for the LLMNR Protocol Similar to the DNS protocol, DNS client computers can use the LLMNR protocol to resolve names on local network segments when DNS servers are unavailable.

The above three types of flows occur frequently in different types of traffic and can be regarded as common background traffic [25,26]. However, they carry less effective information by themselves, which is not conducive to the model's extraction and learning of specific business features of this type of traffic. Therefore, they are filtered out in the data processing.

3.2. Inter-flow features extraction

This section describes the construction method of the network flow relation graph used in the model.

The purpose of constructing the flow relation graph is to obtain the inter-flow features of network flows, specifically whether there is any correlation between flows. Therefore, we use flows as nodes in the graph. In this process, the directionality between flows is not necessary, so we use an undirected graph for construction, where edges represent the existence of correlations between flows. Whether to add an edge between nodes depends on the node attributes. The node attributes we consider are the triplet source IP, destination IP, protocol. The specific process is as follows:

1. Create an empty undirected graph: First, initialize an empty undirected graph G.
2. Add flows as nodes: Iterate through all flows, and for each flow, add it as a node to the graph G. Each node represents a flow and has three attributes: source IP, destination IP, and protocol. Make sure each node has a unique identifier to distinguish different flows.
3. Check flow attributes and add edges: For each flow node A in the graph, iterate through all other flow nodes in the graph (B, C, etc.). For each pair of flow nodes, perform the following checks: if the source IP, destination IP set of flow A has common values with the source IP, destination IP set of flow B, and they use the same protocol, add an edge between node A and node B. This edge represents the association between flow A and flow B.
4. Repeat the checking process: Continue to repeat step 3 for the remaining flow nodes in the graph until all possible pairs of flow nodes have been considered.
5. Complete the flow relationship graph: When all flows have been traversed, and their relationships have been checked and added to the graph, the undirected graph G is completed. This graph represents flows as nodes and their relationships as edges.

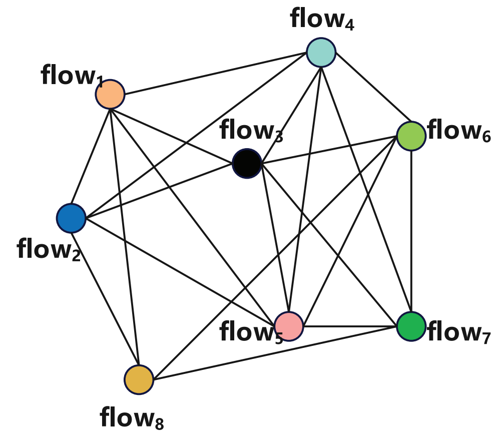


Fig. 2. An example of the flow relation graph.

Algorithm 2: Construct Flow Relation Graph

Input: The set F of flows; Each flow is represented as a triplet $\{\mathrm{S\_IP}, \mathrm{D\_IP}, \mathrm{protocol}\}$.
Output: The flow relation graph G.

Function `ConstructGraph(`F`)`:
  Initialize an empty undirected graph G;
  for *each flow* $\{\mathrm{S\_IP}, \mathrm{D\_IP}, \mathrm{protocol}\}$ *in* F **do**
    Generate a unique node ID using S_IP, D_IP, and protocol;
    Add the node to graph G;
  for *each node* X *in graph* G **do**
    for *each node* Y *in graph* G **do**
      if $X \neq Y$ *and* S_IP *or* D_IP *of* X *is the same as* Y, *and* protocol *is also the same* **then**
  return graph G;

Fig. 2 provides an example of a flow relation graph, which includes 8 nodes representing 8 flows., while the presence of an edge between two nodes indicates that the source IP of one of these two network flows is the same as the source/destination IP of the other flow, or the destination IP of one flow is the same as the source/destination IP of the other flow and they have the same protocol.

3.3. Intra-flow features extraction

We use flow spectrograms as a representation of intra-flow features, while using statistical features as supplements. The following will explain them separately.

3.3.1. Flow time–frequency spectrogram generation

Network traffic data can be considered as a discrete time series, typically analyzed in the time domain in previous studies. This approach, however, may not fully leverage the potential information within the frequency domain, either because it is theoretically unavailable or challenging to utilize when solely focusing on time domain analysis. Conversely, relying solely on frequency domain information may result in the loss of valuable information from the time domain.

Therefore, we use time–frequency spectrograms to depict the features of traffic in both the time and frequency domains. The traffic in the time domain undergoes Fourier transform processing, resulting in the generation of a time–frequency spectrogram for the traffic data. This spectrogram encapsulates the distinctive features of the traffic data in both time and frequency domains, facilitating the comprehensive utilization of information within the traffic.

We employ the short-time Fourier transform (STFT) to generate spectrograms of flows. The STFT offers precise time–frequency resolution within a predetermined window of fixed size. In practical applications, the STFT is frequently computed using overlapping analysis windows. These windows slide in a manner that introduces dependencies between adjacent windows, thereby mitigating the loss of accuracy at boundary locations. Typically, the STFT function can be expressed as

$$\begin{aligned}\mathrm{STFT}\{x[t]\}(m,\omega) &\equiv X(m,\omega)\\ &= \sum_{-\infty}^{+\infty} x[t]\,w[t-mH]\,e^{-j\omega t}\end{aligned} \tag{1}$$

where $x(t)$ denotes the input data sampled at time t, $w[t]$ is the sliding window, ω denotes the phase, m signifies the position of the window, and H is the overlap constant when the window is sliding continuously.

The STFT possesses linearity property, defined by

$$\begin{aligned}\mathrm{STFT}\left\{ax_1[t]+bx_2[t]\right\} &= a\cdot\mathrm{STFT}\left\{x_1[t]\right\}\\ &\quad +b\cdot\mathrm{STFT}\left\{x_2[t]\right\}\end{aligned} \tag{2}$$

In order to avoid complex calculations and preserve the most relevant information, it is common practice to process only the magnitude of the STFT. This results in a joint representation of the time and frequency domains, i.e., spectrogram. It is defined as

$$\mathrm{spectrogram}\{x[t]\}(m,\omega) \equiv |X(m,\omega)|^2 \tag{3}$$

When applied to characterize network traffic, for a flow, we take the first n bytes of it to form a byte sequence and use F as the representation of this flow in the time domain. Considering that the byte sequence is a discrete time series, it is necessary to perform a discrete Fourier transform on each window. In this case, the STFT can be represented as

$$\begin{aligned}\mathrm{STFT}\{\mathrm{F}[i]\}(m,\omega) &\equiv X(m,\omega)\\ &= \sum_{-\infty}^{+\infty} \mathrm{F}[i]\,w[i-mH]\,e^{-j\omega i}\end{aligned} \tag{4}$$

where $\mathrm{F}[i]$ denotes the ith sample point, $1 \le i \le n$. And the spectrogram is

$$\mathrm{spectrogram}\{\mathrm{F}[i]\}(m,\omega) \equiv |X(m,\omega)|^2 \tag{5}$$

The spectrogram can be seen as a two-dimensional visual representation of time and frequency domain information. It simultaneously reflects information from both the time and frequency domains, as opposed to the one-dimensional representation in the time domain.

Here, we examine the flow's spectrogram using information entropy theory. The encrypted flow is treated as a time series, and we apply the sample entropy method to gauge its information content. Sample entropy quantifies the complexity of a time series, indicating the likelihood of new information occurring. It measures the probability of new information in a conditional manner: the more complex the time series, the higher the sample entropy. This metric can effectively reflect the information content in a two-way packet size sequence.

For a sequence of flow bytes of length N,

$$\{b(1), b(2), b(3), \ldots, b(N)\} \tag{6}$$

Let the similarity comparison threshold be r and the subsequence length metric be m. The original sequence is reconstructed so that $(N-m+1)$ subsequences can be obtained as

$$\{X(1), X(2), X(3), \ldots, X(N-m+1)\} \tag{7}$$

Each subsequence is denoted as

$$X(i) = b(i), b(i+1), \ldots, b(i+m-1) \tag{8}$$

For any subsequence $X(i)$ we can define its distance from subsequence $x(j)$ as

$$d_{(ij)} = max\left\{\left|b_i(k)-b_j(k)\right| \,\middle|\, b_i(k)\in X(i), b_j(k)\in X(j)\right\} \tag{9}$$

where $i \le j \le N-m+1, i \ne j$. Calculate the template matching probability of the subsequence $X(i)$:

$$B_i^m(r) = \frac{num\left[d_{ij} < r\right]}{N-m} \tag{10}$$

In turn, the average similarity can be calculated for all subsequences:

$$B^m(r) = \frac{\sum_{i=1}^{N-m+1} B_i^m(r)}{N-m+1} \tag{11}$$

Due to the length of the raw byte sequence of the flow is a finite value, the sample entropy of the sequence is obtained as

$$\mathbf{SampEn}(seq) = \lim_{N\to\infty}\left[-ln\frac{B^{m+1}(r)}{B^m(r)}\right] \tag{12}$$

The analysis indicates that the calculation of sample entropy in the time domain is linear, while the Fourier transform possesses the property of maintaining linear operations. Therefore, following the short-time Fourier transform and considering the introduction of noise, the information entropy in the frequency domain representation of the flow byte sequence does not surpass that of the time domain representation. Consequently, the information entropy of the flow byte sequence in the time–frequency spectrogram representation exceeds that of the time domain representation. From this, we can obtain the following inequality.

$$\mathbf{SampEn}(seq) < \mathbf{tfEn}(seq) \le 2\cdot\mathbf{SampEn}(seq) \tag{13}$$

where $\mathbf{tfEn}(seq)$ represents the entropy of the sequence in the time–frequency domain.

3.3.2. Supplement with statistical features

In the context of network traffic classification, statistical features are utilized to enhance the process of encrypting traffic. Statistical features involve analyzing various characteristics and patterns within the encrypted traffic data. These features may include packet size distribution, inter-arrival time, payload length, and statistical moments such as mean and variance.

By leveraging statistical features, the encrypted traffic can be categorized into different classes or types, allowing for more accurate identification and classification. These features can be extracted from the encrypted traffic data and utilized in machine learning algorithms or statistical models to train and build classifiers.

The utilization of statistical features in encrypting traffic classification provides a robust and reliable method to differentiate various types of encrypted traffic.

We use statistical features as a supplement to spectrograms, and together they form the intra-flow features. They mainly include the following categories:

1. Packet size distribution: Statistics on the size distribution of individual packets in encrypted traffic, including mean packet size, median packet size, mode packet size, standard deviation, percentiles, kurtosis and skewness.
2. Inter-arrival time: Statistics on the time intervals between individual packets in encrypted traffic, such as mean interval, variance and maximum interval.
3. Payload length: Statistics on the length distribution of the payload in encrypted traffic,including forward byte count and packet count, backward byte count and packet count, as well as corresponding maximum, minimum, average, variance, and standard deviation. The payload refers to the actual effective data carried in the packet, excluding the header or encryption parts.
4. Byte sent rate: The rate at which data is sent in bytes over a certain period of time, including the total sent rate, forward sent rate, and back sent rate.
5. Usage of encryption algorithms: The encryption algorithm suites and protocol types used in encrypted traffic.

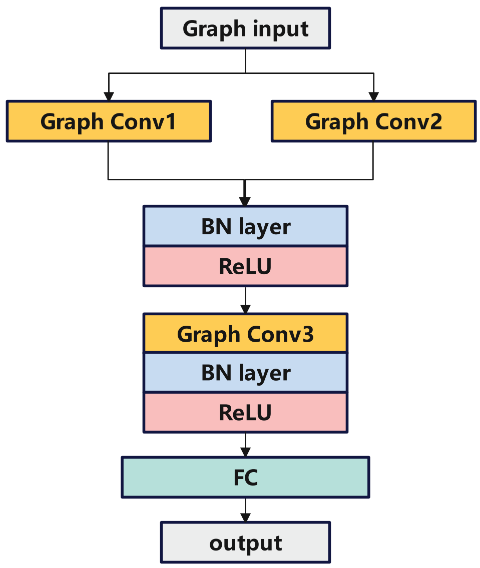


Fig. 3. The framework of inter-flow features extractor.

3.4. Classification module

After obtaining the inter-flow and intra-flow features, we use models that adapt to them respectively for learning, and then complete the final classification using multimodal learning method.

Inter-flow Features Learning We use the GCN model to extract the flow relationship features in flows. GCN is a neural network model specifically designed for processing graph structured data. It performs convolution operations on the graph, effectively capturing local neighborhood information and global topology structure between nodes [27].

The model we used is shown in Fig. 3. It contains 3 graph convolutional layers and one fully connected layer. We use the cross-entropy loss function, and the result of this part of the loss function is noted as $Loss_{Inter}$, and the obtained feature extraction result is M_{Inter}.

$$Loss_{\text{Inter}} = -\sum_{i=1}^{n} p\left(\text{graph}_i\right) \log\left(q\left(\text{graph}_i\right)\right) \tag{14}$$

We use two parallel graph convolutional layers, where these two layers employ different aggregation functions: sum and max, respectively. This approach allows the two graph convolutional layers to capture different feature representations from the graph. It aids the model in capturing a more diverse set of node features, thereby enhancing the model's understanding and representational capacity of the graph.

Intra-flow Features Learning After obtaining the spectrograms representation of the flows, we use CNN to learn them. Spectrograms often contain many local structures, such as changes in frequency and instantaneous changes in time. CNN is capable of automatically learning these local features through convolutional layers. Moreover, by utilizing multiple convolutional and pooling layers, CNN can progressively extract more abstract and higher-dimensional features [28]. Thus, CNN can better capture the complex structures and patterns within spectrograms.

The CNN model we used referred to VGG16. VGG16 is a convolutional neural network model widely used in the field of image processing. This network model has a total of 16 layers, including 13 convolutional layers and 3 fully connected layers. In order to reduce the amount of parameters and computation, we have replaced some convolutional layers with depthwise separable convolution(DSC) layers and reduced the number of layers in the network. All convolutional

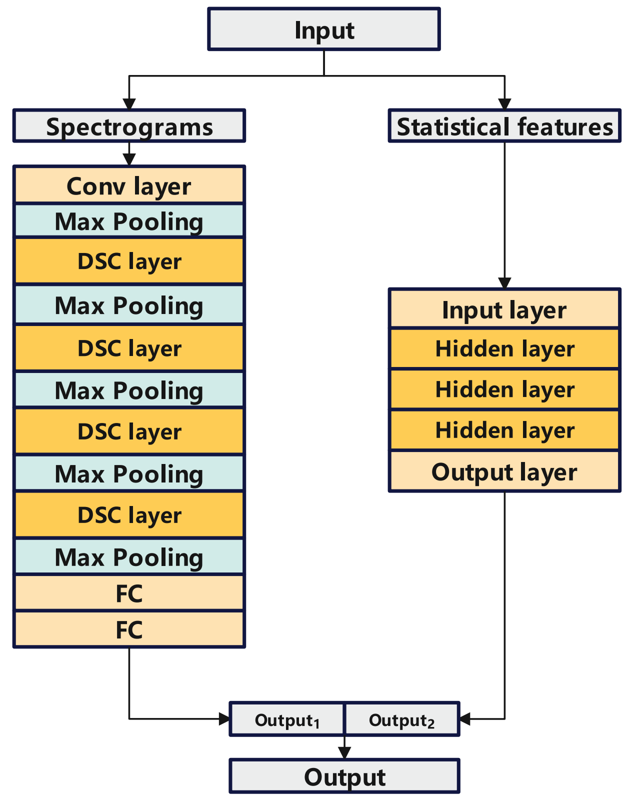


Fig. 4. The framework of intra-flow features extractor.

layers use 3 Convolutional kernel of 3, with a step size of 1 and padding of 1 and all pooling layers are Maximum pooling of 2, with a step size of 2 and no padding.

We use multi-layer perceptron(MLP) to learn statistical features. This MLP model contains three hidden layers. The statistical features of encrypted traffic are obtained by summarizing, calculating, or combining the original features, which may contain complex nonlinear relationships. MLP can automatically learn complex interactions and combination relationships between features through multiple layers of weight combinations and activation functions. It has strong nonlinear modeling ability and can better capture information in the statistical features [29].

The structure of the two model is shown in Fig. 4. For both models, we use cross entropy loss functions defined as, where the final loss function is the sum of the two loss functions, defined as

$$Loss_{\text{CNN}} = -\sum_{i=1}^{n} p\left(spect_i\right) log\left(q\left(spect_i\right)\right) \tag{15}$$

$$Loss_{\text{MLP}} = -\sum_{i=1}^{n} p\left(f_i\right) log\left(q\left(f_i\right)\right) \tag{16}$$

$$Loss_{\text{Intra}} = Loss_{\text{CNN}} + Loss_{\text{MLP}} \tag{17}$$

We use the cross-entropy loss function, and the result of this part of the loss function is noted as $Loss_{Intra}$, and the obtained feature extraction result is M_{Intra}.

Multimodal fusion The outputs of the three sub-models are fused in a cascading manner. Specifically, GCN is used to process the flow relation graph, obtaining the representation vectors for each node as inter-flow features. MLP and CNN are respectively used to process the statistical features and spectrogram of the encrypted traffic. And their output vectors are concatenated to obtain a representation vector. Finally, the two representation vectors are concatenated to obtain the final representation vector, which is input into the fully connected layer for classification.

We combine the previously extracted features M_{Intra} and M_{Inter} for classification through the fully connected layer and softmax layer, also using the cross-entropy loss function, denoted as $Loss_{\mathrm{mf}}$. We want the neural network to utilize both the intra-flow features and inter-flow features so that they can complement each other. In order to achieve the best classification effect of the model, we assume that the final classification effect will be optimal when the inter-flow features extraction module, the intra-flow features extraction module and the classification module all reach the optimal parameters. Therefore, the final loss function $Loss_{all}$ of the whole model is

$$Loss_{\mathrm{all}} = Loss_{\mathrm{Intra}} + Loss_{\mathrm{Inter}} + Loss_{\mathrm{mf}} \tag{18}$$

In multimodal learning, the weights of each modality's loss function represent the importance of that modality. To ensure that intra- and inter-flow features receive equal attention during model training, we set the weights of the loss functions for these two modalities to be the same.

4. Experiment

This section presents the network traffic dataset used for the experiments as well as the implementation details, evaluation metrics, and baseline methodology. Also in this section, encrypted traffic classification experiments are performed for all models, including quantitative evaluation, ablation studies, and sensitivity analysis, to comprehensively and specifically evaluate the performance of MeDF.

4.1. Dataset

We used two publicly available datasets:ISCX VPN-nonVPN 2016 [30] and Malicious_TLS [31]. The former is a commonly used dataset in encrypted traffic recognition research, while the latter is used to evaluate the performance of MeDF in malicious traffic classification. We do not use the entire traffic data for training models when using these two datasets.

The Malicious_TLS dataset includes traffic generated by 22 active malicious code families from 2018 to 2021, as well as some benign traffic. All of this traffic is collected from real networks and encrypted through TLS. We selected six types of malicious traffic and some benign traffic for our experiment. The composition of the dataset is shown in Table 2.

The ISCX VPN-nonVPN dataset comprises 14 categories of regular traffic (non-VPN) and VPN traffic (including VOIP, VPNVOIP, P2P, VPN-P2P, etc.), encompassing various types of traffic such as browsing, VoIP, streaming, chatting, email, and more. The dataset has a size of approximately 28 GB, and for the analysis, six labeled categories of regular traffic were used. The available dataset composition is outlined in Table 3.

In practical scenarios, network traffic may contain various types of data, including malicious traffic and regular traffic. Malicious traffic and regular traffic often have different characteristics and behavioral patterns. Therefore, testing on these two types of traffic can help determine the effectiveness of the model. Additionally, using these two types of traffic for experimentation can evaluate the model's generalization ability. Therefore, testing the model using these two types of traffic can better simulate real-world application scenarios and ensure the model's usability in a real environment.

Table 2
The composition of the Malicious_TLS dataset.

Category	Flow count
Arachni	2000
Awvs	2000
Burpsuite	2000
Shifu	2000
Tiggre	2000
Tor	2000
Benign	4000

Table 3
The composition of the ISCX VPN-nonVPN dataset.

Category	Flow count
Chat	31 334
Mail	24 719
File Transfer	67 790
Streaming	36 128
Torrent	169 309
VoIP	53 040

4.2. Experiment setup

This section describes the configuration of the experimental environment we used, the evaluation metrics and the baseline.

(1) Environment Settings: The computer specifications used in the experimental environment are as follows: Intel Core i7-9700K CPU, Nvidia 3080 GPU, 32 GB RAM, and Windows operating system. The experiments are conducted based on the PyTorch 1.11.0 framework and implemented in the Python language of version 3.7.13.

(2) Evaluation Metrics:In this paper, Accuracy, Recall, False Positive Rate (FPR) and Precision are used to assess the performance of the model and are calculated as follows.

$$\mathrm{Accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{19}$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{20}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{21}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{22}$$

where TP, FP, TN and FN denote true positive, false positive, true negative and false negative, respectively.

(3) Baseline: To better evaluate the performance advantages and disadvantages of the MeDF proposed in this paper, we selected 6 currently well-performing encrypted traffic classification models as, two of which use unimodal learning and the other two use multimodal learning.

1 D-CNN An end-to-end classification method that converts bytes to image grayscale values and then classifies traffic using 1 D-CNN [9].

XGBoost A decision tree model that is an implementation of the Gradient Boosting algorithm and is widely used in various machine learning tasks [5].

ACID A clustering model for encrypted traffic classification [8].

ProGraph A graph neural network based encrypted traffic classification model [13].

AppNet A multimodal deep learning framework that uses the packet length sequence of the initial packet and the payload bytes as input. The former is modeled using Bi-LSTM and the latter using 1D-CNN [6].

MIMETIC A multimodal deep learning framework also using 4 protocol fields extracted from the bi-directional flow and payload bytes of the initial packet as input, the former modeled using GRU (a simplified version of LSTM) and the latter modeled using 1-D CNN [7].

Table 4
Comparison of classification results on Malicious_TLS (%).

Models	Accuracy	Recall	FPR	Precision
1D-CNN	92.37	91.76	0.31	90.78
XGBoost	90.29	90.17	0.37	90.21
ACID	96.13	95.87	0.22	95.98
ProGraph	91.55	91.34	0.33	91.39
AppNet	95.52	95.30	0.27	95.52
MIMETIC	96.73	96.32	0.26	96.29
MeDF (Ours)	98.57	98.62	0.22	98.14

Table 5
Comparison of classification results on ISCX VPN-nonVPN 2016 (%).

Models	Accuracy	Recall	FPR	Precision
1D-CNN	89.68	89.25	0.42	89.17
XGBoost	88.32	88.27	0.41	89.06
ACID	92.73	92.48	0.30	91.63
ProGraph	94.35	94.54	0.30	93.89
AppNet	90.17	90.34	0.34	90.28
MIMETIC	91.56	91.45	0.32	90.89
MeDF (Ours)	94.73	94.56	0.28	93.86

4.3. Comparison of classification results

The comparison of the classification performance of different models is shown in Tables 4 and 5, respectively, for the experimental results on Malicious_TLS and ISCX VPN-nonVPN 2016.

MeDF achieves the best classification results of all the approaches. It can be seen that the multimodal models AppNet, MIMETIC, and MeDF have improved in accuracy compared to the unimodal 1D-CNN model on both datasets. This indicates that the multimodal models are able to learn the information in the encrypted traffic data more fully after obtaining more than one class of feature inputs, which leads to better classification results. Similar to the 1D-CNN model, MeDF also converts the raw byte information of the flow into images and then analyzes them. The difference between 1D-CNN and MeDF is that 1D-CNN converts flows into grayscale spectrograms and uses 1D CNN for analysis, while MeDF converts flows into time–frequency spectrograms and uses 2D convolution for analysis, in addition to adding inter-flow relationship information, thus achieving better results.

In addition, compared to AppNet and MIMETIC models, MeDF has further improvement in classification accuracy, which we consider because MeDF combines both inter-flow features and intra-flow features modalities.MIMETIC uses the first 576 bytes of payload and 4 protocol features as input; AppNet extracts features from the first 1014 bytes of the first packet features from the first 1014 bytes of the first packet, as well as the time-relative features of the packet length sequence. The different modalities used by these two models are essentially inter-flow features. MeDF, on the other hand, uses features embedded in the flow-relations graph to complement the inter-flow features of encrypted traffic in addition to the inter-flow features using the time–frequency graph to characterize it. This indicates that using both intra-flow features and inter-flow features can achieve better complementary effects when using modality fusion methods for encrypted traffic classification, and thus optimize the classification results.

The confusion matrices of MeDF on two datasets for classification are given by Fig. 5. It can be observed that on Malicious_TLS, MeDF achieves close to 1 prediction accuracy across all categories, while on ISCX VPN-nonVPN, the accuracy for each category is almost all below 95%. We attribute this to the more complex data structure of ISCX VPN-nonVPN compared to Malicious_TLS. Additionally, the weaker correlation between flows leads to less prominent inter-flow features, resulting in a decrease in the model's classification accuracy.

(a) Confusion matrices of MeDF on Malicious_TLS

(b) Confusion matrices of MeDF on ISCX VPN-nonVPN 2016

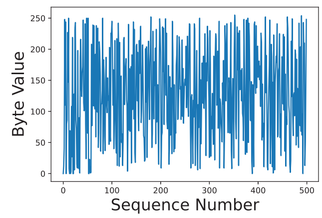
Fig. 5. Confusion matrices of MeDF on two datasets.

4.4. Flow spectrogram

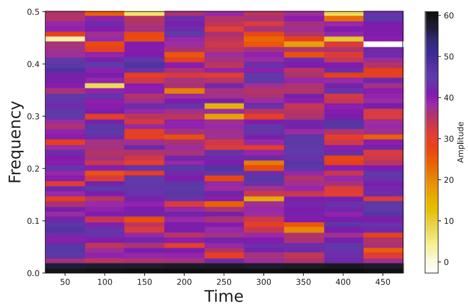
For each flow, we extract the first 500 raw bytes of it as the input of the STFT. For flows less than 500 bytes in length, we pad with 0 to complete 500 bytes. The IP addresses are masked. For the STFT, there are some key parameters that need to be set in advance, including window length, window overlap rate, and window function, etc. The window length we use is 100, with a overlap rate of 67%, and the window function is the Hanning window.

Fig. 6(a) shows the intuitive representation of the original byte stream in the time domain, where the horizontal axis represents time and the vertical axis represents the corresponding bytes. Fig. 6(b) shows the spectrogram, where the horizontal axis represents time, the vertical axis represents frequency.

The spectrogram can be viewed as a two-dimensional visual representation of the joint function of time domain information and frequency domain information of a network flow. Unlike the one-dimensional representation of the network flows in the time domain, the spectrogram representation of the network flows can reflect the

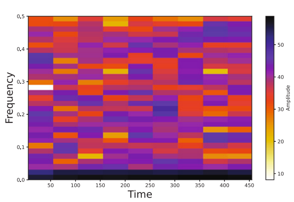


(a) time domain

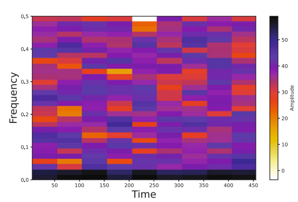


(b) time-frequency domain

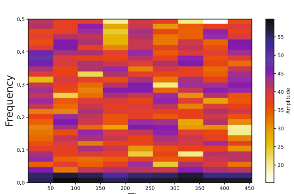
Fig. 6. The representations of flow in different domains.



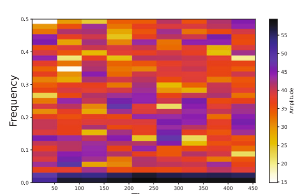
(a) Arachni



(b) Shifu



(c) Tiggre



(d) Tor

Fig. 7. The spectrograms of different applications.

information of the encrypted traffics in both the time and frequency domain.

Spectrogram is a more effective way to characterize network flows than 1D time-series In the STFT process, the features are already "pre-extracted". The amount of information of the features characterized by a single point in the spectrogram is larger than that of a single point in the one-dimensional temporal bits. So it is easier for the classifier to use its information for feature characterization.

Fig. 7 shows the spectrograms of some malicious traffics in the dataset we used. It can be seen that the time–frequency spectrograms of different types of encrypted traffic have more obvious differences, which also confirms the effectiveness of using time–frequency diagrams to characterize the time–frequency domain of encrypted traffic.

Table 6
Ablation study results (%).

Models	Accuracy	Recall	FPR	Precision
MeDF	98.57	98.62	0.22	98.14
intra-MeDF	96.82	92.76	0.31	95.79
inter-MeDF	90.25	89.92	0.97	90.27

Table 7
Complexity of feature extraction.

	Time complexity	Space complexity
Spectrogram	$O(\mathrm{NM}\,log\mathrm{M})$	$O(\mathrm{NM})$
Flow relation graph	$O(\mathrm{V}^2)$	$O(\mathrm{E})$

4.5. Ablation study

In this section, we further illustrate the effectiveness of combining inter-flow features and intra-flow features for multimodal learning by constructing 2 weakened models of MeDF.

We construct two weakened models, one is to remove the inter-flow features extraction module from the original model and use only the intra-flow features extraction module and classification module for classification, called intra-MeDF; the other is to remove the intra-flow features extraction module from the original model and use only the flow relation graph and GCN model for classification, called inter-MeDF. We conducted experiments on the Malicious_TLS dataset and obtained the results as shown in Table 6.

Both aspects of the modality extracted by MeDF are useful. And intra-flow features are better for traffic classification than inter-flow features we used. It can be seen that there is a significant decrease in the classification accuracy of both weakened models compared to MeDF: the accuracy of intra-MeDF decreased by 1.75% compared to MeDF, while the accuracy of inter-MeDF decreased by 8.32% compared to MeDF. The other three indicators of MeDF are also better than the weakened models. It indicates that the inter-flow features and intra-flow features of network traffic are able to complement each other in the modality fusion process.

By comparing intra-MeDF and inter-MeDF, the accuracy, recall and precision of inter-MeDF decreased by 6.57%, 2.84% and 5.52% respectively compared to intra-MeDF. We think that it is because when constructing the flow relation graph, we did not use too much information about the network flows themselves, mainly retaining the correlation features between flows. Therefore, using only flow relation graph for node classification to complete traffic classification is not effective enough. When using intra-flow features to complete classification, it contains more information about the network flow, thus achieving better classification results.

4.6. Complexity analysis

In this section, we analyzed the complexity of MeDF, including time complexity, space complexity, the number of model parameters and run time per-epoch.

(a) Complexity of feature extraction

In this part, we analyze the time and space complexity of feature extraction in MeDF, including the construction of spectrogram and flow relation graph. The result is as shown in the Table 7.

The method we use to construct the spectrogram is the STFT, which involves dividing the original signal into multiple windows and performing Fast Fourier Transform (FFT) on each window. Assuming the length of the original signal is N and the window length is M, the time complexity of FFT is $O(\mathrm{M}\,log\mathrm{M})$. For STFT, since FFT needs to be performed on different time windows, the overall time complexity is $O(\mathrm{NM}\,log\mathrm{M})$. The space complexity depends on the memory required to store the time–frequency spectrograms. For STFT, it typically requires

Table 8
Model complexity.

Model	Parameter size (millions)	RTPE (s)
MeDF (Ours)	2.54	47.5
MIMETIC	1.78	39.4

storing the spectral information for each time window. Therefore, the space complexity can be approximated as $O\,(\mathrm{NM})$.

When constructing a flow relation graph, as we need to iterate over all subsequent flows for each flow, the time complexity can be represented as the total computation of a nested loop. Assuming there are V flows, the first flow needs to iterate over the remaining $(\mathrm{V}-1)$ flows, the second flow needs to iterate over the remaining $(\mathrm{V}-2)$ flows, and so on, until the second-to-last flow needs to iterate over the last flow. Therefore, the total computation can be represented as:

$$[(\mathrm{V}-1)+(\mathrm{V}-2)+\cdots+1] \tag{23}$$

This is the sum of an arithmetic progression, and its sum is $\left(\frac{\mathrm{V}(\mathrm{V}-1)}{2}\right)$. Therefore, the computational complexity is $O\left(\mathrm{V}^2\right)$.

The flow relation graph can be stored in the form of a sparse matrix. The number of edges is much smaller than the square of the number of nodes, resulting in a space complexity of $O\,(\mathrm{E})$, where E is the number of edges in the flow relation graph.

(b) Model complexity

In this part, we analyze the number of model parameters and RTPE of two multimodal learning models. The result is as shown in the Table 8.

It can be seen that MeDF has an increase in both parameter size and RTPE compared to MIMETIC. This is because, in order to obtain more comprehensive feature information, the model of MeDF is more complex than MIMETIC. Due to the significant improvement in classification performance of MeDF, these additional costs are acceptable.

5. Conclusion

In this paper, we propose an encrypted traffic classification model based on multimodal fusion, called MeDF. We mainly consider inter-flow features and intra-flow features when selecting features of network traffic, in order to allow the classification model to learn information about multiple aspects of traffic. We validated the effectiveness of our model on a real-world encrypted traffic dataset and were able to achieve an accuracy of over 98%, superior to multimodal learning methods that use only intra-flow features. This shows that the combination of intra-flow and inter-flow features is effective. Moreover, compared to MeDF using only intra-flow features and only inter-flow features, the accuracy is improved by 2.62% and 9.19% when using both features, respectively. This further illustrates the necessity of using both intra- and inter-flow features for multimodal flow classification.

CRediT authorship contribution statement

Xiangbin Wang: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Qingjun Yuan:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Yongjuan Wang:** Writing – review & editing, Validation, Methodology. **Gaopeng Gou:** Methodology, Validation, Writing – review & editing. **Chunxiang Gu:** Methodology, Validation. **Gang Yu:** Methodology, Formal analysis. **Gang Xiong:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in the article are all publicly available data and have been cited.

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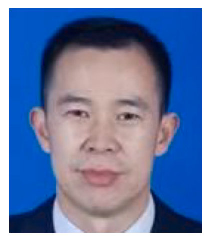

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